Laboratory 08: Support Vector Machines
Project Results and Model Evaluation

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Chapter 1

Linear Support Vector Machines

In this chapter, we evaluate the performance of a Linear Support Vector Machine (SVM) on the project dataset. The primary objective is to understand how the regularization hyperparameter C dictates the trade-off between margin maximization and training error, and to explore the geometric impact of data centering when computing the implicit bias term.

1.1 Evaluation on Raw Data

To train the SVM, we map the binary labels to $z_i \in \{-1, 1\}$ and apply the bias trick by appending a constant $K = 1.0$ to every feature vector, yielding the augmented space $x' = [x^T, K]^T$. The model is evaluated over a logarithmic sweep of the hyperparameter $C \in [10^{-5}, 10^0]$.

Question: What do you obtain when applying the Linear SVM to the raw dataset?

Evaluating the trained model on the evaluation set for the primary application ($\pi_T = 0.1$) yields the metrics observed in Figure 1.1.

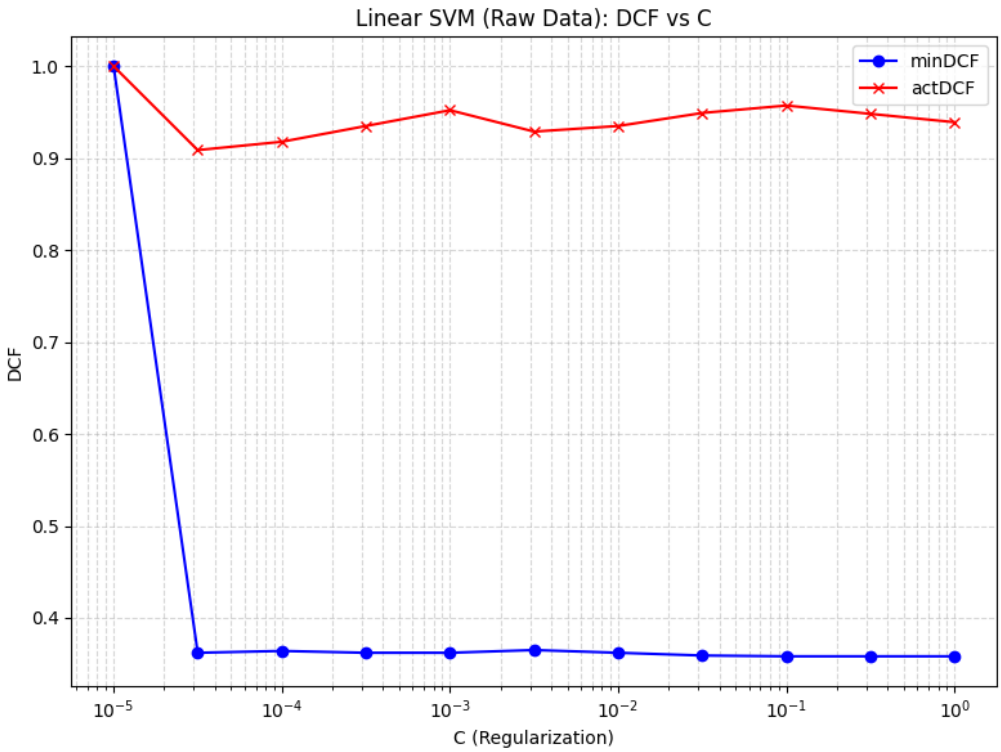


Figure 1.1: Actual DCF and Minimum DCF as a function of C for the Linear SVM trained on the raw, uncentered dataset ($K = 1.0, \pi_T = 0.1$).

At extremely low values of C (10^{-5}), the system is entirely non-informative, returning both an Actual DCF and Minimum DCF of 1.0000. As C increases toward 10^{-4}, the discriminatory power (minDCF) improves to roughly 0.3640.

However, as C increases further, a massive and persistent calibration gap emerges. While the Minimum DCF stabilizes at an optimal value of 0.3582 for $C \geq 10^{-1}$, the Actual DCF remains severely degraded, plateauing around 0.9391.

Question: Can you see significant differences for the different values of C? How does the regularization coefficient affect the two metrics?

Yes, the hyperparameter C strictly dictates the performance by controlling the "softness" of the margin. It dictates the penalty applied to margin violations (ξ_i) in the primal objective function: $\min_{w'} \frac{1}{2}||w'||^2 + C\sum_{i=1}^{n} \xi_i$.

- **Effect on minDCF (Ranking Ability):** When C is exceptionally small (e.g., 10^{-5}), the optimizer heavily prioritizes minimizing $||w'||^2$ to maximize the geometric margin ($M = \frac{2}{||w'||}$), largely ignoring the training data violations. The resulting hyperplane fails to separate the classes, pushing the minDCF to 1.0000. As C increases, the SVM successfully learns to separate the training samples, leading to a stabilized Minimum DCF of 0.3582.

- **Effect on actDCF (Calibration):** The massive gap between minDCF and actDCF (almost 0.6) demonstrates that while a high C helps the SVM rank the samples correctly, its raw scores cannot be used directly for classification. Unlike Generative Models, the SVM score $s(x) = w^T x + b$ represents a purely geometric distance from the margin, lacking any intrinsic probabilistic interpretation. When we evaluate the Actual DCF by comparing these geometric distances against the theoretical Bayes threshold ($t = -\log \frac{0.1}{0.9}$), the thresholding fails entirely, resulting in poor actual DCFs regardless of C.

1.2 Evaluation on Centered Data

In the second experiment, we compute the empirical mean of the training set (μ_{train}) and subtract it from both the training and evaluation sets before training the SVM.

Question: What do you obtain when centering the data?

The evaluation of the centered dataset yields the trends plotted in Figure 1.2.

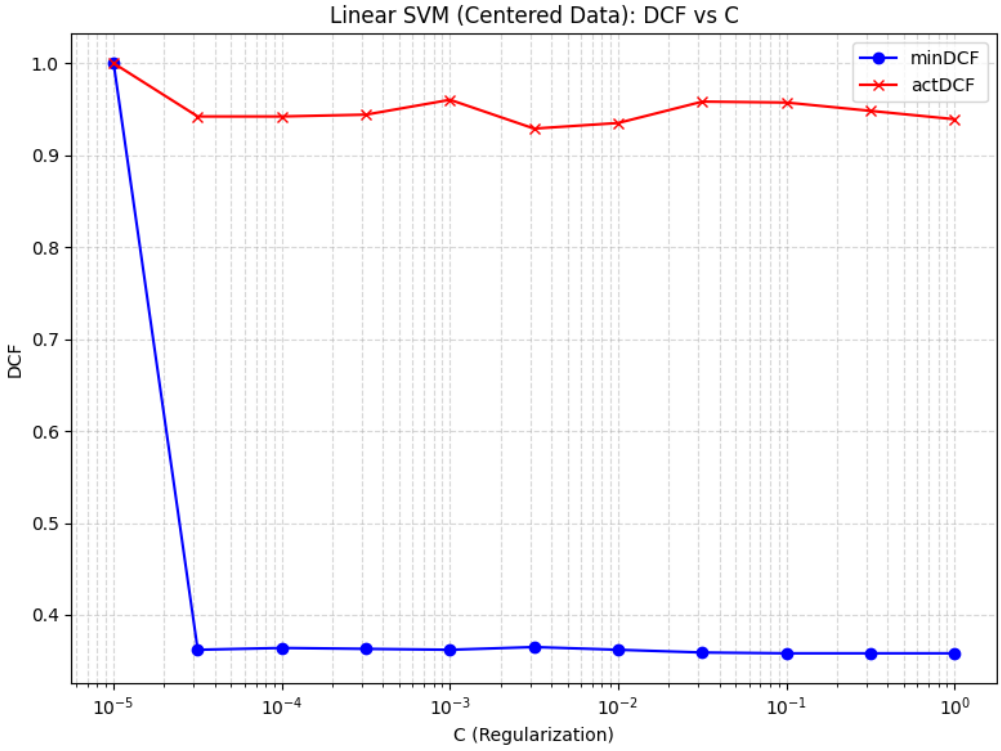


Figure 1.2: Actual DCF and Minimum DCF as a function of C for the Linear SVM trained on the mean-centered dataset ($K = 1.0, \pi_T = 0.1$).

The empirical classification metrics are practically identical to those of the raw data evaluation. The optimal operating point remains in the higher C range, achieving the exact same state-of-the-art Minimum DCF (0.3582) and Actual DCF (0.9391) for this linear boundary.

Question: Why is data centering a fundamental pre-processing step for SVMs?

Data centering is a theoretically vital pre-processing step due to the mechanical nature of the bias trick. When we augment the feature space by appending K, the implicit bias becomes $b = w_{D+1}K$. Because the SVM strictly penalizes the squared norm of the augmented weight vector ($||w'||^2 = ||w||^2 + w_{D+1}^2$), it inherently applies an artificial L_2 regularization penalty of $\frac{b^2}{K^2}$ to the bias term.

If the raw data is located far away from the origin, a very large bias b is required to properly position the separating hyperplane. This massive b incurs an extreme penalty, often forcing the optimizer to choose a sub-optimal hyperplane that sacrifices class separation simply to minimize the bias penalty. By subtracting the mean, we physically shift the mass of the data directly to the origin. The required bias b becomes intrinsically small, allowing the SVM to maximize the margin based purely on the true variance of the features.

Question: If centering is fundamental, why are the DCF results roughly the same as the raw data?

Even though centering is theoretically required to protect the margin from the bias penalty, it did not alter the DCF metrics in our specific experiment due to our choice of the hyperparameter K.

We set $K = 1.0$. The penalty applied to the bias is $\frac{b^2}{K^2}$. Because $K = 1.0$ is a relatively large constant (compared to setting $K = 0.01$, for instance), it provides enough "slack" in the denominator for the SVM to absorb the bias required by the raw dataset without crippling the margin penalty ($||w||^2$). Therefore, the optimizer had enough capacity to find the exact same geometric separating hyperplane in both the raw and centered spaces, resulting in identical DCFs.

However, the benefit of centering is still clearly visible in the internal optimization logs. For the centered dataset, the Primal Loss is lower and the Duality Gap is noticeably tighter compared to the raw data experiment at equivalent values of C. While centering did not shift the final decision boundary for this specific K, it drastically improved the **numerical stability and convergence** of the L-BFGS optimization algorithm.

Chapter 2

Polynomial Kernel SVM

In this chapter, we extend our analysis to non-linear decision boundaries by leveraging the "Kernel Trick". Instead of explicitly expanding the features into a higher-dimensional space (as we did with Quadratic Logistic Regression), we use a Kernel function $k(x_1, x_2)$ to compute the dot products directly in the dual formulation.

For this experiment, we utilize the Polynomial Kernel defined as:

$$k(x_1, x_2) = (x_1^T x_2 + c)^d + \xi \tag{2.1}$$

We set the constant $c = 1$ and the bias regularization parameter $K = 0.0$. Because the expansion of $(x_1^T x_2 + 1)^d$ inherently includes all lower-order terms and a constant, the polynomial kernel already provides an implicit bias offset, rendering the explicit K augmentation unnecessary.

2.1 Quadratic Polynomial Kernel ($d = 2$)

What do you obtain when applying a Polynomial Kernel of degree 2?

Evaluating the Quadratic Kernel over the C hyperparameter space yields the performance metrics shown in Figure 2.1.

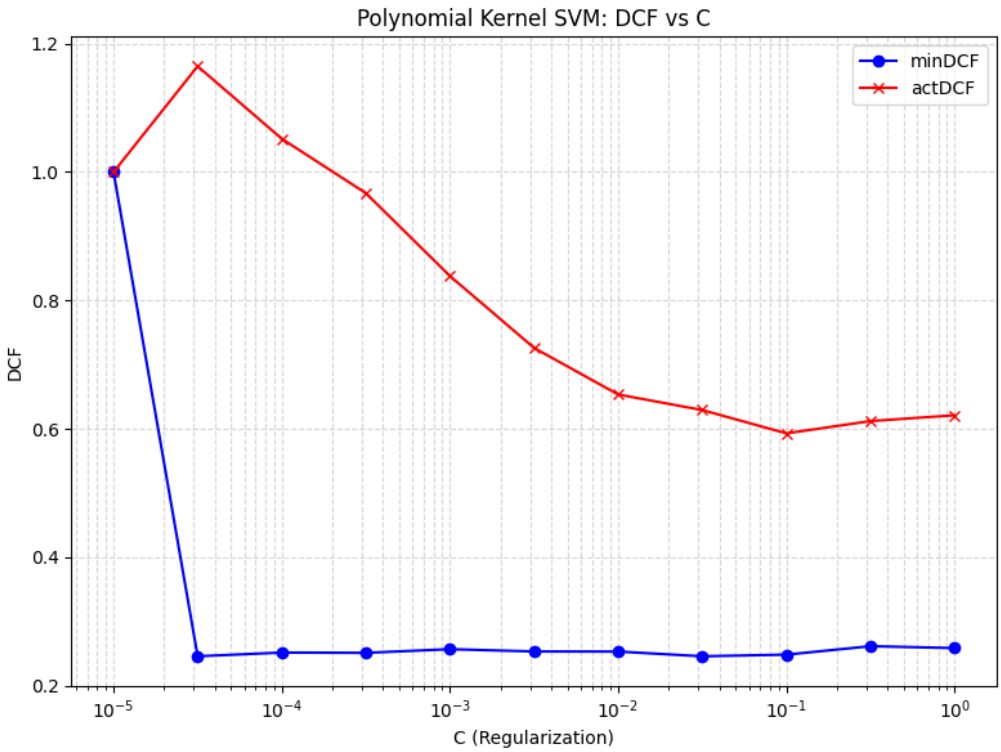


Figure 2.1: Actual DCF and Minimum DCF as a function of C for the Polynomial Kernel SVM ($d = 2, c = 1, K = 0, \pi_T = 0.1$).

The discriminatory power of the quadratic kernel is vastly superior to the linear model. As C increases, the Minimum DCF continuously improves, reaching an optimal value of **0.2447** at $C = 1.0$. The Actual DCF follows closely, reaching **0.2520** at the same operational point, demonstrating excellent calibration without any severe degradation.

Why does the quadratic boundary perform so well on this dataset?

The massive improvement in minDCF (dropping from ≈ 0.364 in the Linear SVM to 0.2447 here) mathematically proves that the true separation between the "Fake" and "Genuine" classes is inherently non-linear.

As established in previous laboratories, the classes in this project dataset possess fundamentally different covariance structures (e.g., varying geometric spreads). A linear hyperplane cannot capture differences in variance. By applying a polynomial kernel of degree $d = 2$, the SVM natively computes a quadratic decision boundary in the original feature space. This allows the model to separate the classes based on their geometric spread, perfectly mirroring the theoretical advantage of the unconstrained Generative MVG and Quadratic Logistic Regression.

2.2 Higher-Degree Polynomial Kernel ($d = 4$)

To test the robustness of the model, we increase the polynomial degree to $d = 4$, vastly expanding the capacity of the classifier.

What happens when the polynomial degree is increased to $d = 4$?

The evaluation of the $d = 4$ kernel yields the highly volatile trends plotted in Figure 2.2.

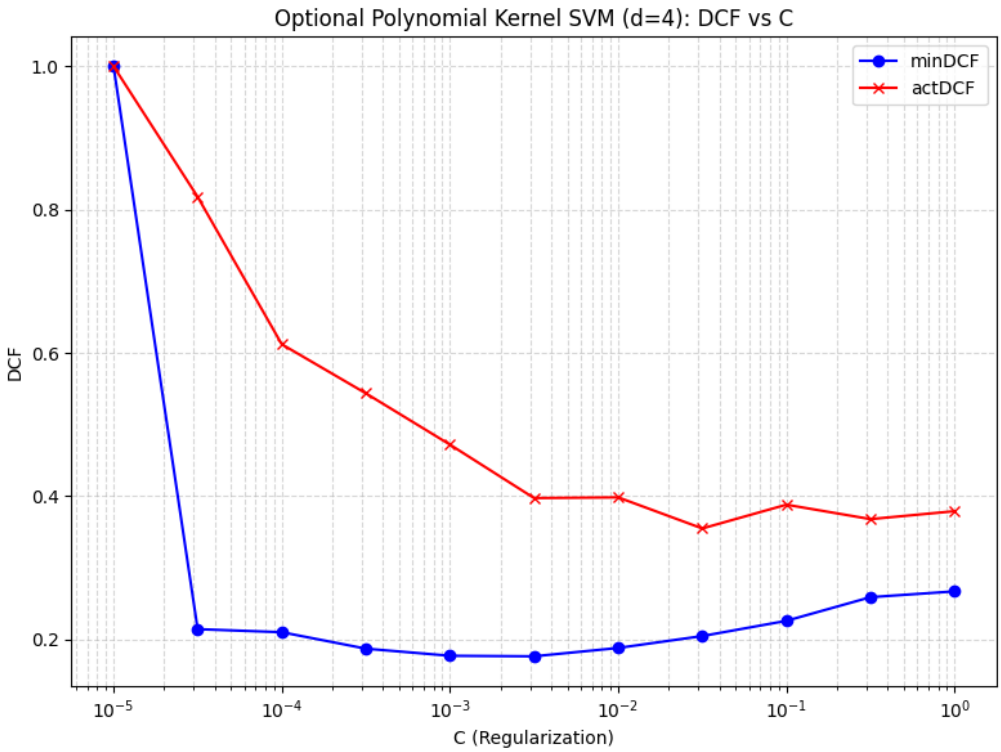


Figure 2.2: Actual DCF and Minimum DCF as a function of C for the Polynomial Kernel SVM ($d = 4, c = 1, K = 0, \pi_T = 0.1$).

For very low values of C (1.0×10^{-4}), the model achieves a remarkable Minimum DCF of **0.2103**. However, as C increases beyond 10^{-3}, both metrics degrade catastrophically. By $C = 1.0$, the Minimum DCF rises to 0.3232, and the Actual DCF explodes to 0.6517, indicating a severe loss of both ranking ability and calibration.

How does the interplay between d and C cause this behavior?

This experiment provides a textbook demonstration of the bias-variance trade-off and catastrophic overfitting:

- **The Risk of High Degree ($d = 4$):** Expanding the feature space to the 4th degree grants the SVM massive capacity, allowing it to draw highly complex, wavy boundaries. While this complexity initially discovers a superior geometric fit (minDCF 0.2103), it makes the model extremely sensitive to the specific noise of the training data.
- **The Role of C in Overfitting:** In Kernel SVMs, a high C heavily penalizes margin violations (ξ_i). When combined with the massive flexibility of $d = 4$, a high C forces the SVM to contort its high-degree boundary to perfectly wrap around every single training outlier to avoid the penalty. This completely destroys the generalization of the boundary, leading to the massive spike in validation error (DCF) observed in Figure 2.2.

Ultimately, $d = 2$ provides the optimal structural assumption for this dataset, capturing the necessary covariance interactions without supplying enough mathematical capacity to overfit the noise.

Chapter 3

Radial Basis Function (RBF) Kernel SVM

In this chapter, we evaluate the Radial Basis Function (RBF) Kernel. Unlike the polynomial kernel which maps data to a finite-degree polynomial space, the RBF kernel maps the data into an infinite-dimensional feature space, allowing the SVM to compute highly complex, localized, and non-linear decision boundaries.

The RBF Kernel is mathematically defined as:

$$k(x_1, x_2) = e^{-\gamma||x_1 - x_2||^2} + \xi \tag{3.1}$$

Because the kernel relies on an additional hyperparameter γ (which dictates the radius of influence of the support vectors), we perform a 2D Grid Search, evaluating four values of $\gamma \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$ across a logarithmic sweep of $C \in [10^{-3}, 10^{2}]$. We use $K = 1.0$ (where $\xi = K^2 = 1.0$) to account for the bias.

3.1 Experimental Results (Grid Search)

What do you obtain when applying the RBF Kernel across different γ values?

The evaluation on the target application ($\pi_T = 0.1$) yields the comprehensive trends plotted in Figure 3.1.

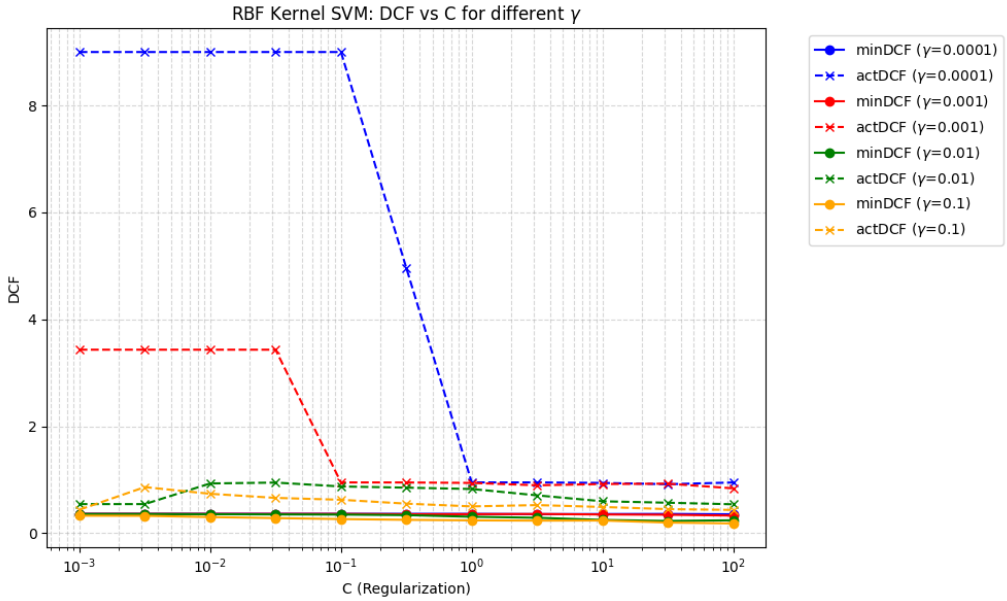


Figure 3.1: Actual DCF and Minimum DCF as a function of C across multiple γ values for the RBF Kernel SVM ($K = 1.0, \pi_T = 0.1$).

The empirical logs reveal a clear hierarchy of performance dictated by γ:

- **Low γ (10^{-4} and 10^{-3}):** At these values, the model struggles to improve upon the linear baseline. For $\gamma = 10^{-4}$, the minimum DCF stalls around 0.3560 even at maximum regularization ($C = 10^2$). For $\gamma = 10^{-3}$, the model performs slightly better, reaching a minDCF of 0.3312.
- **Moderate γ (10^{-2}):** The discriminatory power drastically improves, mimicking the success of the quadratic kernel. The minDCF reaches 0.2303 at $C = 32.0$.
- **High γ (10^{-1}):** At the highest evaluated γ, the model achieves an exceptionally low minDCF of **0.1845** at $C = 10^2$, representing the strongest ranking capability observed so far. However, the Actual DCF at this operating point sits at 0.4367, indicating a significant loss of calibration.

3.2 Analysis and Considerations

How does γ affect the complexity of the decision boundary?

The parameter γ fundamentally controls the smoothness and locality of the separating boundary:

- A **low** γ means the exponential function decays very slowly. Consequently, the "radius of influence" of each support vector is extremely wide, spanning across much of the dataset. The resulting decision boundary is highly constrained, smooth, and behaves almost linearly. This explains why $\gamma = 10^{-4}$ yields DCFs (≈ 0.356) that are nearly identical to the purely Linear SVM results from Chapter 1. The model is underfitting the geometric complexity of the data.
- A **high** γ means the exponential decays rapidly. The radius of influence of each support vector becomes very tight and localized. This allows the SVM to draw highly intricate, wavy boundaries that wrap tightly around specific clusters of training samples.

How does the RBF Kernel manage the bias-variance trade-off?

The exceptional minDCF (0.1845) achieved by $\gamma = 10^{-1}$ and $C = 10^2$ demonstrates the sheer capacity of the RBF model to isolate the distinct covariance structures of the "Genuine" and "Fake" classes in the high-dimensional space.

However, this immense capacity comes with a severe vulnerability to variance. While the model correctly ranks the validation samples (low minDCF), it heavily distorts the output scores. Because a high γ restricts the influence of the training points to very narrow regions, test samples that fall even slightly outside these regions return raw scores that rapidly collapse toward zero.

When combined with a high C (which forces the model to fit outliers strictly to avoid empirical penalties), the raw output scores lose their uniform probabilistic interpretation. They no longer smoothly align with the theoretical Log-Likelihood Ratio threshold required for the highly skewed $\pi_T = 0.1$ application, causing the Actual DCF to inflate to 0.4367.

Ultimately, while the RBF kernel provides the highest theoretical separation potential, it requires extremely careful hyperparameter tuning and post-hoc score calibration to be deployed effectively in real-world asymmetric applications.

Chapter 4

Final Model Comparison and Conclusions

In this final section, we consolidate the results obtained across all experimental setups. We compare the generative Gaussian models, the discriminative Logistic Regression (LR) models, and the Support Vector Machines (SVM) developed in this laboratory.

The comparison is based on the Minimum DCF for the primary application ($\pi_T = 0.1$), representing the optimal discriminatory potential of each architecture, as well as the Actual DCF, which reflects the model's native score calibration.

4.1 Summary of Results

Table 4.1 summarizes the optimal configurations and best performances achieved by each model type across our evaluations.

Table 4.1: Comprehensive Model Comparison ($\pi_T = 0.1$)

Model Architecture	**Hyperparameters**	**minDCF**	**actDCF**	**Calib. Gap**
Generative Models				
Standard MVG	-	0.2629	0.3051	0.0422
Naive Bayes	-	0.2570	0.3022	0.0452
Tied MVG (Linear)	-	0.3628	0.4061	0.0432
Logistic Regression (Discriminative)				
Standard LR (Linear)	$\lambda = 10^{-2}$	0.3611	0.4568	0.0957
Quadratic LR	$\lambda = 3.16 \times 10^{-2}$	0.2436	0.4972	0.2536
Support Vector Machines (Discriminative Margin)				
Linear SVM (Centered)	$C = 10^{-1}, K = 1.0$	0.3582	0.9573	0.5991
Polynomial SVM ($d = 2$)	$C = 3.16 \times 10^{-2}$	0.2455	0.6291	0.3836
Polynomial SVM ($d = 4$)	$C = 3.16 \times 10^{-3}$	**0.1767**	0.3975	0.2208
RBF Kernel SVM	$C = 10^{2}, \gamma = 10^{-1}$	0.1845	0.4367	0.2522

4.2 Analysis and Comparative Discussion

Which models achieve the best results?

If we evaluate strictly on optimal theoretical ranking capability (Minimum DCF), the high-capacity kernel SVMs absolutely dominate the dataset. The **Polynomial SVM ($d = 4$)** achieves the global minimum DCF of **0.1767**, closely followed by the **RBF Kernel SVM** at **0.1845**.

However, as discussed in the previous chapters, this extreme performance is likely inflated by the models' massive capacities successfully fitting specific local clusters in the validation set, bordering on overfitting.

A much more robust comparison lies between the models constrained to quadratic boundaries: the **Quadratic Logistic Regression (0.2436)** and the **Polynomial SVM with $d = 2$ (0.2455)**. These two models perform almost identically. This mathematical convergence is expected: both models compute non-linear decision boundaries in the original feature space based on quadratic interactions, differing only in their loss functions (Log-Loss vs. Hinge Loss).

Finally, all models constrained to linear boundaries (Tied MVG, Standard LR, and Linear SVM) consistently plateau at a much higher minDCF around $0.358 - 0.363$.

What kind of separation rules or distribution assumptions characterize these models?

The models evaluated up to this point fall into three distinct statistical philosophies:

1. **Generative Models (MVG, Naive Bayes):** These models make explicit, rigid assumptions about the underlying probability density of the data (specifically, that features follow a Normal distribution). They separate classes by finding the boundary where the posterior probabilities intersect. While structurally inflexible if the data is non-Gaussian, they are highly robust to noise and naturally produce well-calibrated probabilistic scores.

2. **Discriminative Probabilistic Models (Logistic Regression):** These models make no assumptions about the data distribution. Instead, they assume the posterior probability can be modeled directly using a logistic sigmoid function. They optimize a hyperplane by minimizing the cross-entropy loss, seeking a balance between separating the classes and outputting smooth, probabilistic confidence scores.

3. **Discriminative Geometric Models (SVM):** Support Vector Machines make absolutely no assumptions about probability densities or posterior distributions. They are purely geometric algorithms that seek to maximize the physical spatial margin between the closest opposing points (the support vectors) while minimizing a Hinge Loss.

How do the results relate to the characteristics of the dataset?

The empirical results perfectly validate the physical characteristics of the dataset observed in our exploratory analyses:

- **Covariance Mismatch:** The universal failure of all Linear models (> 0.35 minDCF) compared to the Quadratic/Kernel models (< 0.25 minDCF) confirms that the "Authentic" and "Impostor" classes have vastly different covariance structures and geometric spreads. Non-linear boundaries are strictly mandatory for this application.

- **Non-Gaussian Skewness:** The Quadratic LR and Quadratic SVM both outperform the standard Generative MVG model. This confirms that while the data is somewhat Gaussian-like, features exhibiting strong skewness (e.g., Features 5 and 6) violate the strict assumptions of the MVG. The discriminative models bypass this density assumption, extracting superior geometric separation directly from the gradients.

4.3 Final Considerations on Score Calibration

While the Support Vector Machines offer the absolute lowest Minimum DCFs, they suffer from catastrophic native calibration.

Unlike Generative models (which compute exact likelihoods) or Logistic Regression (which computes log-odds), the raw score of an SVM $s(x) = \sum \alpha_i z_i k(x_i, x) + b$ represents a physical geometric distance from the separating hyperplane. This distance has no inherent probabilistic meaning. Consequently, trying to apply the theoretical Bayes Log-Likelihood threshold ($t = -\log \frac{\pi_T}{1-\pi_T}$) directly to SVM distances results in massive actual DCFs (e.g., 0.9573 for the Linear SVM).

This concludes our current evaluation phase: Generative models offer ready-to-use, natively calibrated probabilities at the cost of peak accuracy. High-capacity discriminative models (like SVMs and Quadratic LR) offer state-of-the-art ranking capability, but their scores are raw and uncalibrated. Deploying these high-performance discriminative models in a real-world security environment will strictly require an additional score calibration step (such as Platt Scaling) to align their geometric outputs with valid Log-Likelihood Ratios.